

DATABASE ERD & RELATIONAL DESIGN

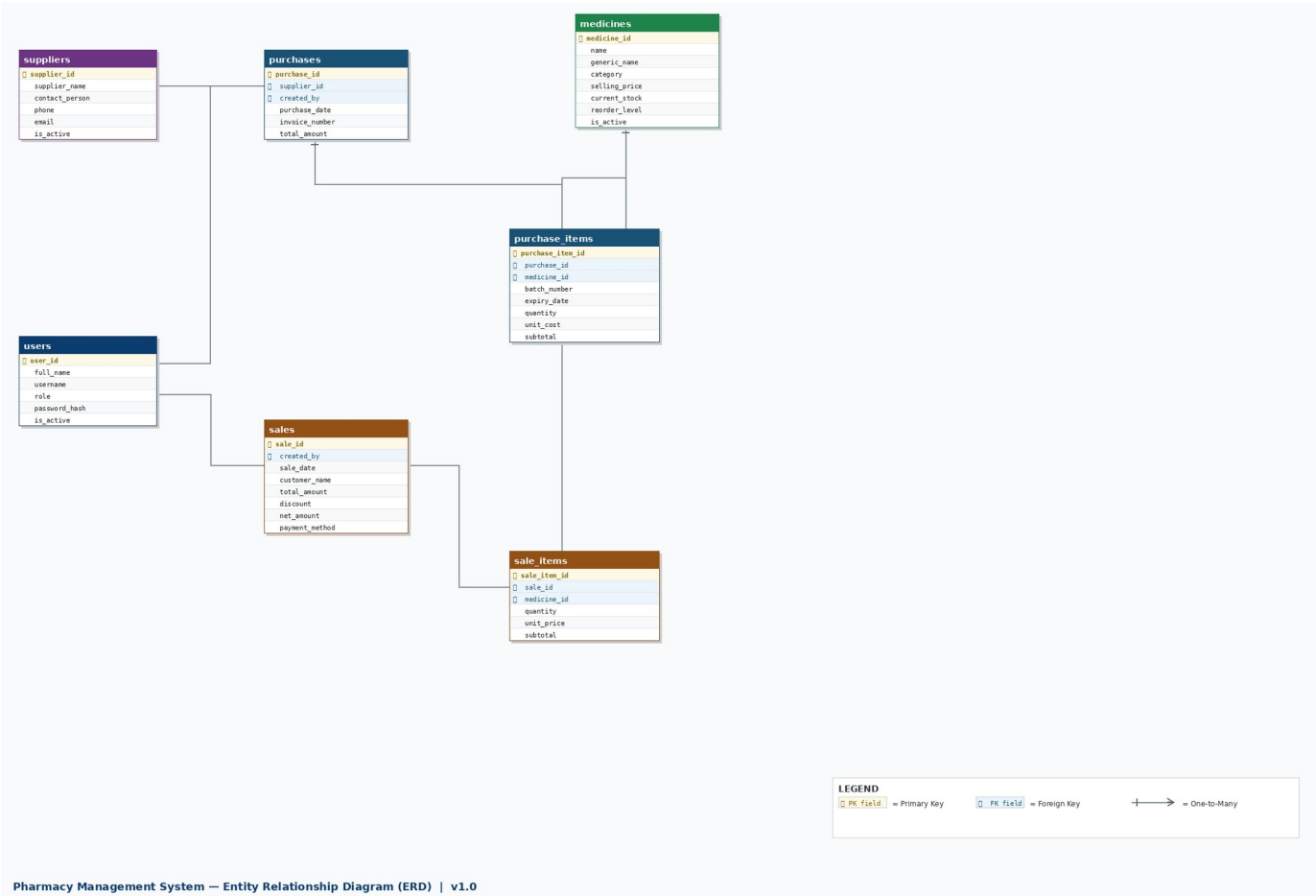
Pharmacy Management System

Entity Relationship Diagram | Relational Schema | v1.0 | March 2026

Database	PostgreSQL 15+ / MySQL 8.0+
Total Tables	7 tables
Normalization	Third Normal Form (3NF) / BCNF
All Relationships	7 FK relationships — all One-to-Many (1:N)

1. Entity Relationship Diagram (ERD)

The diagram below shows all 7 entities, their attributes, and One-to-Many relationships using Crow's Foot notation. Gold rows = Primary Key (PK), Blue rows = Foreign Key (FK). Relationship lines use || (one) and >< (many) crow's foot symbols.



Pharmacy Management System — Entity Relationship Diagram (ERD) | v1.0

Figure 1 — Pharmacy Management System ERD (Crow's Foot Notation, 7 Tables)

2. Relationship Summary

All 7 relationships are One-to-Many (1:N). The table below captures the parent, child, foreign key column, and the referential integrity action on delete.

Parent Table			Child Table	Foreign Key	On Delete Action
suppliers	1	→ N	purchases	supplier_id	RESTRICT (preserve history)
users	1	→ N	purchases	created_by	SET NULL
purchases	1	→ N	purchase_items	purchase_id	CASCADE (delete lines)
medicines	1	→ N	purchase_items	medicine_id	RESTRICT (preserve history)
users	1	→ N	sales	created_by	SET NULL
sales	1	→ N	sale_items	sale_id	CASCADE (delete lines)
medicines	1	→ N	sale_items	medicine_id	RESTRICT (preserve history)

3. Full Relational Schema

Complete column definitions for all 7 tables. Primary Keys highlighted in gold, Foreign Keys in blue.

Table: users

Column	Data Type	Key	Notes
user_id	SERIAL	PK	Auto-increment surrogate PK
full_name	VARCHAR(150)		NOT NULL
username	VARCHAR(80)	UQ	NOT NULL — unique login identifier
email	VARCHAR(150)		NULL allowed
password_hash	VARCHAR(255)		NOT NULL — bcrypt hash, min factor 12
role	VARCHAR(50)		NOT NULL — admin / pharmacist / viewer
is_active	BOOLEAN		DEFAULT TRUE — soft-disable accounts
last_login	TIMESTAMP		NULL — updated on each login
created_at	TIMESTAMP		DEFAULT NOW()
updated_at	TIMESTAMP		DEFAULT NOW()

Table: medicines

Column	Data Type	Key	Notes
medicine_id	SERIAL	PK	Auto-increment surrogate PK
name	VARCHAR(200)		NOT NULL — indexed for search
generic_name	VARCHAR(200)		NULL allowed
category	VARCHAR(100)		NULL — e.g. Antibiotic, Analgesic
unit	VARCHAR(50)		NOT NULL — tablet / syrup / injection
manufacturer	VARCHAR(150)		NULL allowed
selling_price	NUMERIC(10,2)		NOT NULL CHECK >= 0
reorder_level	INTEGER		NOT NULL DEFAULT 10 — triggers low-stock alert
current_stock	INTEGER		NOT NULL DEFAULT 0 CHECK >= 0 — INDEXED
is_active	BOOLEAN		DEFAULT TRUE — soft delete
created_at	TIMESTAMP		DEFAULT NOW()
updated_at	TIMESTAMP		DEFAULT NOW()

Table: suppliers

Column	Data Type	Key	Notes
supplier_id	SERIAL	PK	Auto-increment surrogate PK
supplier_name	VARCHAR(200)		NOT NULL — indexed
contact_person	VARCHAR(150)		NULL allowed
phone	VARCHAR(30)		NULL allowed
email	VARCHAR(150)		NULL allowed
address	TEXT		NULL allowed
city	VARCHAR(100)		NULL allowed
is_active	BOOLEAN		DEFAULT TRUE — soft delete
created_at	TIMESTAMP		DEFAULT NOW()
updated_at	TIMESTAMP		DEFAULT NOW()

Table: purchases

Column	Data Type	Key	Notes
purchase_id	SERIAL	PK	Auto-increment surrogate PK

Column	Data Type	Key	Notes
supplier_id	INTEGER	FK	FK → suppliers.supplier_id — NOT NULL, RESTRICT
created_by	INTEGER	FK	FK → users.user_id — NULL allowed, SET NULL
purchase_date	DATE		NOT NULL DEFAULT CURRENT_DATE — INDEXED
invoice_number	VARCHAR(100)		NULL — supplier invoice reference
total_amount	NUMERIC(12,2)		NOT NULL DEFAULT 0.00 CHECK >= 0
notes	TEXT		NULL allowed
created_at	TIMESTAMP		DEFAULT NOW()

Table: purchase_items

Column	Data Type	Key	Notes
purchase_item_id	SERIAL	PK	Auto-increment surrogate PK
purchase_id	INTEGER	FK	FK → purchases.purchase_id — CASCADE
medicine_id	INTEGER	FK	FK → medicines.medicine_id — RESTRICT
batch_number	VARCHAR(100)		NOT NULL — supplier batch/lot number
expiry_date	DATE		NOT NULL — INDEXED for expiry alert queries
quantity	INTEGER		NOT NULL CHECK > 0
unit_cost	NUMERIC(10,2)		NOT NULL CHECK >= 0
subtotal	NUMERIC(12,2)		NOT NULL — quantity × unit_cost
created_at	TIMESTAMP		DEFAULT NOW()

Table: sales

Column	Data Type	Key	Notes
sale_id	SERIAL	PK	Auto-increment surrogate PK
created_by	INTEGER	FK	FK → users.user_id — SET NULL on user delete
sale_date	DATE		NOT NULL DEFAULT CURRENT_DATE — INDEXED
customer_name	VARCHAR(150)		NULL — optional customer name
customer_phone	VARCHAR(30)		NULL — optional customer phone
total_amount	NUMERIC(12,2)		NOT NULL DEFAULT 0.00
discount	NUMERIC(10,2)		NOT NULL DEFAULT 0.00 CHECK >= 0

Column	Data Type	Key	Notes
	)		
net_amount	NUMERIC(12, 2)		NOT NULL — total_amount minus discount
payment_method	VARCHAR(50)		NOT NULL DEFAULT 'cash'
notes	TEXT		NULL allowed
created_at	TIMESTAMP		DEFAULT NOW()

Table: sale_items

Column	Data Type	Key	Notes
sale_item_id	SERIAL	PK	Auto-increment surrogate PK
sale_id	INTEGER	FK	FK → sales.sale_id — CASCADE
medicine_id	INTEGER	FK	FK → medicines.medicine_id — RESTRICT
quantity	INTEGER		NOT NULL CHECK > 0
unit_price	NUMERIC(10, 2)		NOT NULL — captured at time of sale
subtotal	NUMERIC(12, 2)		NOT NULL — quantity × unit_price
created_at	TIMESTAMP		DEFAULT NOW()

4. Referential Integrity & Business Rules

4.1 Delete Behaviour

- CASCADE: Deleting a purchase also deletes all its purchase_items. Deleting a sale also deletes all its sale_items.
- RESTRICT: Cannot delete a supplier with existing purchases or a medicine with any recorded purchase/sale lines — historical data is protected.
- SET NULL: Deleting a user preserves all their sale and purchase records, with created_by set to NULL.

4.2 Check Constraints

- medicines.current_stock >= 0 — stock cannot be negative.
- medicines.selling_price >= 0 — prices cannot be negative.
- purchase_items.quantity > 0 and sale_items.quantity > 0 — line items must have at least 1 unit.
- purchase_items.expiry_date >= parent purchase date — no already-expired stock on arrival.
- sales.discount >= 0 AND <= sales.total_amount — discount cannot exceed total.

4.3 Transaction-Enforced Rules

- On purchase save: medicines.current_stock += sum of all purchase_item quantities (atomic).
- On sale confirm: Pre-check stock >= requested qty AND expiry valid. Then medicines.current_stock -= sum of sale_item quantities (atomic).
- Any failure triggers a full ROLLBACK — no partial records are ever persisted.

5. Normalization Compliance

Form	Requirement	Status in PMS Schema
1NF	Atomic values; no repeating groups	All line items separated into dedicated tables (purchase_items, sale_items)
2NF	No partial dependency on composite PK	All PKs are single-column surrogates — no composite PKs
3NF	No transitive dependency	All non-key columns depend only on the PK of their table
BCNF	Every determinant is a candidate key	No hidden functional dependencies — schema is fully BCNF compliant

— End of ERD & Relational Design Document —